

Choose an efficient calculation method

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Choose an efficient method for $3,998 + 2,507$ and calculate. Copy or complete the first step.

2. Choose an efficient method for $6,300 \div 25$.

3. Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.

4. Miro says: Miro believes the longest written method is always the safest choice. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Choose an efficient method for $3,998 + 2,507$ and calculate. Copy or complete the first step.

Answer Compensation: $4,000 + 2,507 - 2 = 6,505$.

2. Choose an efficient method for $6,300 \div 25$.

Answer Multiply by 4 then divide by 100, or use known multiples: 252.

3. Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.

Answer A written method is efficient; the product is 339,572.

4. Miro says: Miro believes the longest written method is always the safest choice. Is this correct? Explain.

Answer An efficient method fits the numbers and still allows a reliable check.

Choose an efficient calculation method

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Choose an efficient method for $6,300 \div 25$.

2. Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.

3. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.

4. Identify and correct this misconception: Miro believes the longest written method is always the safest choice.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Choose an efficient method for $6,300 \div 25$.

Answer Multiply by 4 then divide by 100, or use known multiples: 252.

2. Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.

Answer A written method is efficient; the product is 339,572.

3. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.

Answer Answers vary. The justification should refer to number structure and accuracy.

4. Identify and correct this misconception: Miro believes the longest written method is always the safest choice.

Answer An efficient method fits the numbers and still allows a reliable check.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Choose an efficient calculation method

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.

2. Explain why this reasoning fails: Miro believes the longest written method is always the safest choice.

3. Create a new example that tests the same idea as: Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Create one calculation best solved mentally and one best solved using a written method. Justify both choices.
Answer Answers vary. The justification should refer to number structure and accuracy.
2. Explain why this reasoning fails: Miro believes the longest written method is always the safest choice.
Answer An efficient method fits the numbers and still allows a reliable check.
3. Create a new example that tests the same idea as: Would you use mental or written calculation for $7,382 \times 46$? Explain and solve.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.